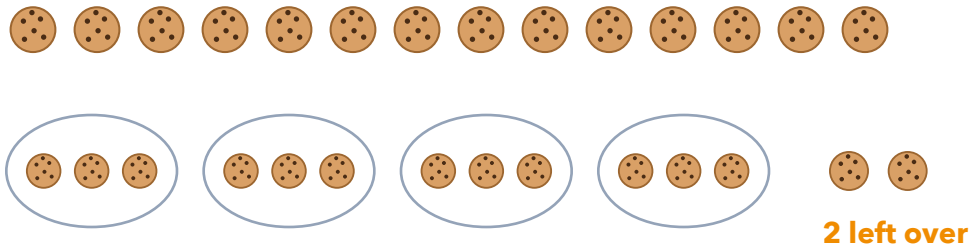
Division with Remainders

Share fairly. What is left over is the remainder.

14 cookies on 4 plates



14 ÷ 4 = 3 R 2 3 on each plate, remainder 2. Check: 4 × 3 + 2 = 14.

Use a times fact: the biggest one that fits

23 ÷ 5:

5 × 4 = 20 (5 × 5 = 25 is too big)

23 − 20 = 3

23 ÷ 5 = 4 R 3

What does the remainder mean?

14 kids, 4 in each car: 3 cars are full and 2 kids still need a ride, so 4 cars.

14 cookies shared by 4 friends: each gets 3, and 2 are left.

Your turn!

3 × 5 = 15

17 ÷ 3 = 5 R 2

For teachers

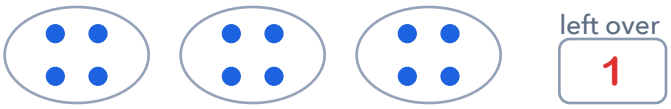
Goal: divide with a remainder and decide what it means in a story (4.OA.3). Share real objects first. Say "What is the biggest times fact that fits? What is left?" The remainder is always less than the divisor.

Division with Remainders: Practice

Share fairly. What is left over?

Part 1 Share on plates Share the dots on the plates. Write what is left over.

1)



left over: 1

13 ÷ 3 = 4 R 1

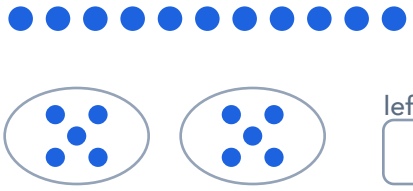
2)



left over: 2

17 ÷ 5 = 3 R 2

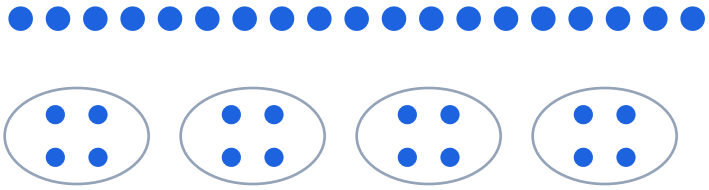
3)



left over: 1

11 ÷ 2 = 5 R 1

4)



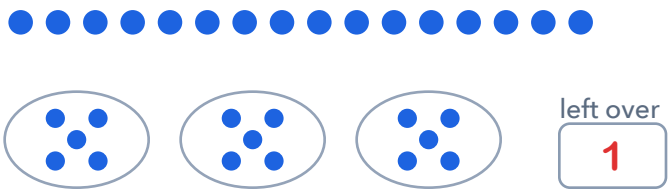
left over: 3

19 ÷ 4 = 4 R 3

Division with Remainders: Practice (continued)

Part 1 Share on plates (continued)

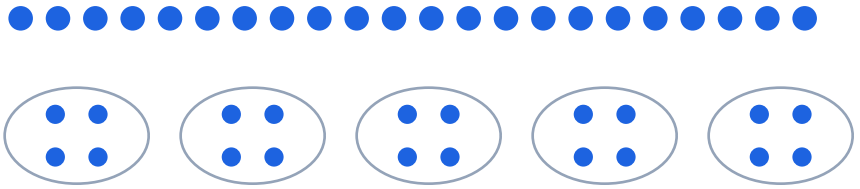
5)



left over: 1

16 ÷ 3 = 5 R 1

6)



left over: 2

22 ÷ 5 = 4 R 2

Part 2 Make groups Circle the groups. Count the leftovers.

7) Circle groups of 4.



3 groups of 4, 2 left over

8) Circle groups of 3.



5 groups of 3, 2 left over

Division with Remainders: Practice (continued)

Part 2 Make groups (continued)

9) **Circle groups of 5.**



4 groups of 5, 3 left over

10) **Circle groups of 2.**



6 groups of 2, 1 left over

11) **Circle groups of 6.**



3 groups of 6, 2 left over

12) **Circle groups of 4.**



4 groups of 4, 3 left over

Part 3 Use a times fact Find the biggest fact that fits, then what is left.

13) **29 ÷ 4: the biggest fact up to 29**

4 × 7 = 28

29 − 28 = 1

so 29 ÷ 4 = 7 R 1

14) **38 ÷ 5: the biggest fact up to 38**

5 × 7 = 35

38 − 35 = 3

so 38 ÷ 5 = 7 R 3

Division with Remainders: Practice (continued)

Part 3 Use a times fact (continued)

15) 45 ÷ 7: the biggest fact up to 45

7 × 6 = 42

45 − 42 = 3

so 45 ÷ 7 = 6 R 3

16) 26 ÷ 3: the biggest fact up to 26

3 × 8 = 24

26 − 24 = 2

so 26 ÷ 3 = 8 R 2

17) 53 ÷ 6: the biggest fact up to 53

6 × 8 = 48

53 − 48 = 5

so 53 ÷ 6 = 8 R 5

18) 61 ÷ 8: the biggest fact up to 61

8 × 7 = 56

61 − 56 = 5

so 61 ÷ 8 = 7 R 5

Part 4 Divide No help this time.

19) 17 ÷ 5 = 3 R 2

20) 23 ÷ 4 = 5 R 3

21) 31 ÷ 6 = 5 R 1

22) 44 ÷ 7 = 6 R 2

23) 19 ÷ 2 = 9 R 1

24) 38 ÷ 9 = 4 R 2

Division with Remainders: Practice (continued)

Part 4 Divide (continued)

25) 27 ÷ 8 = 3 R 3

26) 50 ÷ 6 = 8 R 2

27) 35 ÷ 4 = 8 R 3

28) 47 ÷ 5 = 9 R 2

29) 62 ÷ 7 = 8 R 6

30) 29 ÷ 3 = 9 R 2

Part 5 Bigger numbers No help this time.

31) 73 ÷ 8 = 9 R 1

32) 58 ÷ 9 = 6 R 4

33) 66 ÷ 7 = 9 R 3

34) 85 ÷ 9 = 9 R 4

35) 79 ÷ 8 = 9 R 7

36) 52 ÷ 6 = 8 R 4

Division with Remainders: Practice (continued)

Part 6 What is left?

37) 39 ÷ 5 = 7 R 4

38) 15 ÷ 7 = 2 R 1

39) 22 ÷ 4 = 5 R 2

40) 11 ÷ 7 = 1 R 4

41) 70 ÷ 9 = 7 R 7

42) 25 ÷ 6 = 4 R 1

43) 36 ÷ 7 = 5 R 1

44) 53 ÷ 8 = 6 R 5

45) 17 ÷ 6 = 2 R 5

46) 87 ÷ 9 = 9 R 6

47) 23 ÷ 6 = 3 R 5

48) 38 ÷ 6 = 6 R 2

Division with Remainders: Practice (continued)

Part 6 What is left? (continued)

49) 59 ÷ 8 = 7 R 3

50) 32 ÷ 6 = 5 R 2

51) 41 ÷ 9 = 4 R 5

52) 29 ÷ 6 = 4 R 5

53) 65 ÷ 8 = 8 R 1

54) 43 ÷ 5 = 8 R 3

55) 38 ÷ 7 = 5 R 3

56) 29 ÷ 4 = 7 R 1

57) 43 ÷ 7 = 6 R 1

58) 27 ÷ 4 = 6 R 3

59) 18 ÷ 5 = 3 R 3

60) 59 ÷ 6 = 9 R 5

Division with Remainders: Practice (continued)

Part 6 What is left? (continued)

61) $27 \div 6 = 4$ R 3

62) $32 \div 9 = 3$ R 5

63) $43 \div 8 = 5$ R 3

64) $33 \div 8 = 4$ R 1

65) $14 \div 4 = 3$ R 2

66) $42 \div 9 = 4$ R 6

67) $66 \div 8 = 8$ R 2

68) $33 \div 6 = 5$ R 3

69) $18 \div 8 = 2$ R 2

70) $60 \div 7 = 8$ R 4

71) $41 \div 6 = 6$ R 5

72) $60 \div 9 = 6$ R 6

Division with Remainders: Practice (continued)

Part 7 Remember? Problems from earlier packets.

73) $43 \times 6 =$ **258**

74) 7,281 to the nearest thousand = **7,000**

75) $6{,}304 - 2{,}857 =$ **3,447**

76) in 3,519 the 5 is worth **500**

77) each picture = 10: 9 pictures = **90**

78) 7 m by 4 m: perimeter = **22** m

Part 8 Word problems What does the remainder mean in the story?

79) 26 kids go on a trip. Each van holds 6 kids. How many vans do they need?

Number sentence **26 ÷ 6 = 4 R 2, so 5 vans**

Answer **5** vans

Draw or show your work

80) Mia shares 30 stickers equally with 4 friends. How many stickers does each friend get?

Number sentence **30 ÷ 4 = 7 R 2, 7 each**

Answer **7** stickers

Draw or show your work

81) Ben puts 45 eggs in cartons of 12. How many eggs are left over?

Number sentence **45 ÷ 12 = 3 R 9**

Answer **9** eggs

Draw or show your work

Division with Remainders: Practice (continued)

Part 8 Word problems (continued)

82) A ribbon is 50 inches long. Kim cuts pieces 8 inches long. How many whole pieces can she cut?

Number sentence 50 ÷ 8 = 6 R 2, 6 pieces

Answer 6 pieces

Draw or show your work

83) Challenge: 58 books go on shelves that hold 9 each. How many shelves are full, and how many books are left?

Number sentence 58 ÷ 9 = 6 R 4

Answer 6 R 4

Draw or show your work

Check-In: Division with Remainders

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Make groups

1) Circle groups of 3.



6 groups of 3, 1 left over

Part 2 Use a times fact

2) 14 ÷ 3: the biggest fact up to 14

3 × 4 = 12

14 − 12 = 2

so 14 ÷ 3 = 4 R 2

3) 17 ÷ 4: the biggest fact up to 17

4 × 4 = 16

17 − 16 = 1

so 17 ÷ 4 = 4 R 1

Part 3 Divide

4) 18 ÷ 4 = 4 R 2

5) 21 ÷ 4 = 5 R 1

6) 77 ÷ 9 = 8 R 5

7) 73 ÷ 9 = 8 R 1

8) 11 ÷ 3 = 3 R 2

9) 51 ÷ 9 = 5 R 6